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ACADEMIC APPOINTMENTS

National University of Singapore (NUS)

04/2014-Present

ACADEMIC

Professor at NUS Business School

11/2024-Present

Associate Professor at NUS Business School

01/2017-10/2024

Co-Director, NUS Master of Science in Business Analytics (MSBA) **01/2017-Present**

- Managed the NUS MSBA program operations and strategy.
- Secured more than **S\$ 2.5 million industry scholarship funding** for NUS MSBA students.
- Led the MSBA industry capstone internship program and secured over 100 industry capstone projects annually.
- Mentored students in international and local competitions.
- Provided mentorship to student start-ups at NUS Incubator.

Visiting Associate Professor at NUS School of Computing

04/2015-04/2017

- Led industry collaboration efforts in Healthcare and IT sectors for BAC.
- Established a new Healthcare Analytics specialization for the NUS MSBA program.
- Mentored student start-ups at NUS Incubator.

ADMINISTRATIVE

Co-Director, NUS Business Analytics Centre (BAC)

01/2017-Present

- Managed daily operations and strategic planning at BAC.
- Secure **S\$ 5.52 million industry and research funding**.
- **Built a vibrant industry community for BAC with over 100 partners**, including Accenture, AXA, Alibaba, DBS, EY, Grab, Huawei, IBM, Johnson & Johnson, Morgan Stanley, SAP, Shopee, TikTok, and more.
- Established the Industry Advisory Board for BAC, comprising of prominent industry leaders from various sectors.
- Enhanced BAC oversea collaboration with leading academic institutes and industry partners.

Assistant Dean (Industry Relations) at NUS Business School

04/2023-06/2025

Dean's Fellow at NUS Business School

04/2023-06/2025

- Strengthened the collaboration with external partners to foster mutually beneficial relationships.
- Built a strategic partnership framework to facilitate engagement with industry partners.

- Provided guidance to faculty members on effective strategies for industry collaboration and engagement.

INDUSTRY PROFESSIONAL APPOINTMENTS

IBM Corporation

04/2009-04/2017

Chief Architect (Client Technical Advisor) of IBM AP Public Sector 03/2013-04/2017

- Led the design of IBM Business Analytics Reference Architecture (BARA) and provided technical leadership for key mega deal engagements in the ASEAN public sector, including projects such as the Singapore National Electronic Health Record (NEHR), Singapore Customs New TradeNet and TradeXchange, MSF SSNet.
- Spearheaded the technical solution design of the **first worldwide Watson Cognitive Computing Platform for citizen service**.
- Played a pivotal role as a core member of an IBM GMU (Growth Market Unit) Distinguish Engineer (DE) Challenge project, focusing on the DHL Logistics Cloud. Designed a global logistics operation and optimization solution for DHL.
- Led the IBM GMU technical Community in ASEAN and developed solution architecture strategy for the public sector.

Lead Architect and Senior Manager in IBM R&D Labs

05/2011-03/2013

- Led the analytics and optimization development and service activities at IBM China Labs, overseeing a team of 150+ developers, engineers, and architects.
- Led the solution design for **the NingBo Logistics Cloud project (USD 33M)**, which was the **first IBM Industry Cloud** deployment in the Greater China Group (GCG).
- Spearheaded IBM's worldwide Optimization Cloud strategy and development initiatives.
- Managed the analytics & optimization service team, providing consultancy and solutions in supply chain/logistics, manufacturing, retail, and financial service industries. Projects included RICHTeK supply chain planning, Shanghai Electric Group Company (SEC) production planning solution, Hongkong International Terminals (HIT) intelligent yard planning, China Foreign Exchange Trade System (CFETS) transaction optimization solution, ATM Restocking Cash Management optimization solution, among others.

Advisory Architect in IBM Software Labs

04/2009-04/2011

- Managed the ILOG optimization service and solution team at IBM Shanghai Lab.
- Led the IBM first optimization-based SCM solution win (MPO – Manufacturing Productivity Optimization at FAW Car) in GCG, overseeing engagement, solution design, and end-to-end project delivery. (This project achieved **USD 20+ million in annual production cost saving** for one of the largest automotive companies in China and was showcased at the IBM IMPACT 2010 conference.)
- Spearheaded the solution design of various anchor customer projects, including Bank of QingDao (BQD) SOA project, NOL Empty Container Reposition project, Chery CPM project, AnHui tobacco production planning, TAL production planning, DACHSER transportation management system.
- Established and chaired IBM GCG BAO (Business Analytics and Optimization) community, leading the BAO team in designing and developing cross-brand solution assets based on IBM ILOG Optimization, Cognos, and SPSS software framework.

Motorola Electronics Pte Ltd, Singapore**04/2007-03/2009**

Research Scientist in Motorola Labs responsible for providing technology solutions and consultancy to the Motorola Integrated Supply Chain (ISC) organization. Key projects include:

- Developed an optimization-based Surface Mount Technology (SMT) lines scheduling system for Motorola mobile manufacturing sites, resulting in a 15% improvement in SMT line throughput and a reduction in scheduling lead time from hours to minutes.
- Developed an inventory optimization model to optimize Motorola Owned Inventory (MOI) and Vendor Managed Inventory (VMI) in warehouses, leading to a 20% reduction in overall inventory.
- Developed a vehicle routing optimization solution for Motorola Singapore's inbound supply chain operations, resulting in a 133% improvement in lead time reduction and cost savings of 28%.

Micron Technology Inc (Tech Semiconductor), Singapore**05/2006-03/2007**

Senior Staff Engineer in the Manufacturing Science section responsible for designing planning and scheduling solutions based on optimization technology and queuing theory for the advanced 300mm DRAM wafer FAB. Key achievements include:

- Reduced DRAM wafer fabrication cycle time from about 110 days to around 70 days, resulting in a significant improvement in production efficiency and work in process (WIP) inventory reduction.
- Increased wafer FAB throughput by 3-5%.

EDUCATION

National University of Singapore (NUS), Singapore**06/2002-09/2006**

Joint with Massachusetts Institute of Technology (*MIT*), under Singapore-MIT Alliance (SMA) program

Ph.D., Manufacturing System and Technology, September 2006

Zhejiang University, Hangzhou, China**07/1995-03/2002**

M.Eng, Manufacturing System, March 2002

B.Eng, Mechanical Engineering, June 1999

BOARD MEMBERS AND PUBLIC SERVICE APPOINTMENTS

- Member of SAP Asia Pacific and Japan Academic Advisory Board, 01/2024 - Present
- Council Member, China Cloud System Pioneer Strategic Alliance (中国云体系产业创新战略联盟), 11/2018 – Present
- Advisor, FingerDance, 01/2023 - Present
- Advisor, Pioneer Semiconductor Research, 10/2022–09/2024
- Singapore Tourism Board (STB) Data Advisory Panel, 02/2020–01/2023
- Member of NUS Furnace Incubator Advisory Committee, 01/2016–12/2023
- Advisor, Neptune Logistics, 03/2021–02/2022
- Chief Technical Advisor, JINVOVO Technology Pte Ltd, 09/2019 -08/2020
- Advisor, IBD Technology Pte Ltd, 09/2018- 08/2019
- Advisor, Smart City Research Institute (China, GuangXi), 11/2018–10/2019

- Council Member, Singapore Blockchain Technology Foundation, 05/2018 – 04/2019
- Co-opted Member, Singapore Computer Society, Business Analytics Chapter, 01/2018– 01/2019
- Advisor, eBeeCare Pte Ltd, 12/2015– 11/2017
- Member, Luxembourg Institute of Science and Technology (LIST) Strategy Planning Committee, 06/2017– 12/2017
- Member, IBM AP Invention Disclosure Review Board, 01/2015– 04/2017
- Member, IBM Analytics and Optimization Architecture Board, 05/2012– 12/2013
- Member, IBM Software Defined Environment Architecture Board, 03/2013 – 03/2014
- Chairman, IBM Business Analytics and Optimization Association, 01/2010 – 12/2012
- Member, IBM Shared University Research program review board, 01/2012 – 12/2012
- Member, IBM-China Ministry of Education collaboration board, 01/2011 – 12/2011
- Member, IBM China Development Labs architecture board, 01/2010– 12/2013

HONORS AND AWARDS (Selected)

- [INFORMS UPS George D. Smith Prize](#) (NUS MSBA Programme) 04/2025
- [Gartner 2023 Supply Chain Award for Customer or Patient Innovation \(eZTracker Project\)](#) collaborating with Zuellig Pharma) 05/2023
- [Leading Academic Data Leaders 2021](#) by Chief Data Officer (CDO) Magazine 01/2021
- IBM Invention Plateau Award 11/2015
- IBM Invention Leadership Award 10/2015
- Master Certified Architect in the Open Group (Open CA) 06/2014
- IBM Expert Certified Architect 04/2014
- IBM Patent Award 06/2013
- IBM Outstanding Technical Achievement Award (the highest recognition for technical staff in IBM) 11/2011
- IBM Outstanding Employee Equity Award 11/2012
- Best Paper Award at 2010 IBM GCG RTE Technical Conference 10/2010
- IBM Customer Service Star 10/2010
- IBM Eminence and Excellence Award 05/2010
- Motorola Performance Management (PM) Excellent Award 01/2009
- Nominee for the Andrew Fraser Prize 2007 (best postgraduate paper), Institute of Mechanical Engineers (IMechE) 04/2007
- Singapore Government Scholarship 2002-2006
- Outstanding Master of Zhejiang Province (top 1%) 03/2002
- Outstanding Bachelor of Zhejiang University (top 5%) 06/1999
- Distinguished Student Scholarship in Zhejiang University 1995-2002

RESEARCH INTERESTS

Trustworthy AI, Blockchain, Business Analytics,
Modelling and Optimization in Supply Chain, Manufacturing System

PATENTS AND INVENTION DISCLOSURES

1. Yan Pang, Peihua Mai, Wei Liu, Xianglong Du, “A content safety filtering system and method for large language models”, CN 202610726575.4, Filed in May 2026

2. Yan Pang, Peihua Mai, Wei Liu, Yingqing Yang, “A Scalable and Accurate Privacy-Preserving Inference Method and System for Large Language Models”, CN 202610726576.9, Filed in May 2026
3. Yan Pang, Ran Yan, Peihua Mai, “Method And Apparatus For Privacy-Preserving Large Language Models (LLMs) Inference”, 2023049, Filed in Jan 2024
4. Yan Pang, Ran Yan, “A Dynamic Risk Awareness And Assessment System, Apparatus And Method For Digital Assets”, 202311553089.X, Filed in Nov 2023
5. Yan Pang, Xhoni Shollaj, “A Device And Method For Generating Marketing Content For A Subject”, 202310626287.8, Filed in May 2023
6. Yan Pang, “Method and Apparatus for Privacy-Preserving Data Sharing”, 202111467585.4, CN 116226062 A, Filed in Dec 2021, (**Licensed to Chongqing Fei Xiang Industrial Internet Co., LTD** <https://www.feiiplat.com/>)
7. Yan Pang, “System Component Failure Diagnosis”, [US9916194 B2](#), Granted in Mar, 2018
8. J.W. Chen, M. Donovang-Kuhlisch, R. Faferko, Y.X. Liu, Yan Pang, “Application Service Aggregation and Management”, [US9760952 B2](#), Granted in Sep, 2017. (**The Cloud Service Broker designed in this patent has been incorporated in IBM Cloud Computing product offerings:** <https://www.ibm.com/services/cloud/multicloud>)
9. Yan Pang, J. Xu, “System and Method for Intelligent Advertisement using CAPTCHAs”, CN820130769, Granted in May, 2015
10. Yan Pang, “Method and System for Building Mathematical Optimization Model using Interactive Graphical Pipeline”, IPCOM000237716D, Published in Oct, 2014
11. H.T. Peng, Yan Pang, Y. Zhang, L. Nie, Y.Z. Cui, “System and Method of Smart Traffic Control based on Prediction”, IPCOM000236690D, Published in Apr, 2014
12. Yan Pang, Y.Z. Cui, and Philip Starhill, “New format for mathematical optimization model”, IPCOM000220104D, Published in Apr, 2012 (The ALP optimization file format in IBM ILOG CPLEX Studio is from this invention disclosure)
13. “Method and System for Printed Circuit Board Production Scheduling”, by Yan Pang, Y.S. Yip and H. Zhang, Motorola Invention Disclosure CML07329, Nov, 2008
14. “Evaluation and Benchmarking Method for Surface Mount Technology Assembly Line”, by Yan Pang, K.T. Kee, Motorola Invention Disclosure CML07341, Dec, 2008

PUBLICATIONS (selected)

Conference Papers (* Corresponding author)

1. Peihua Mai, Zhuoyan Shao, Xinbao Qiao, Meng Zhang, Xinyue Zhou*, Yan Pang*, “Illusory Pattern Perception Drives Spurious Inference in Large Language Models”, Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026), accepted.
2. Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang*, “When Sample Selection Bias Precipitates Model Collapse”, Forty-third International Conference on Machine Learning (ICML 2026, Main Conference) [\[arXiv\]](#)
3. Peihua Mai, Xuanrong Gao, Youlong Ding, Xianglong Du, Wei Liu, Yan Pang*, “SharedRequest: Privacy-Preserving Model-Agnostic Inference for Large Language Models”, Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), Volume 1: Long Papers, pp. 7129–7150, 2026. [\[arXiv\]](#) [\[Paper\]](#)
4. Youlong Ding, Peihua Mai, Jingqi Zhang, Sherman S. M. Chow, Minxin Du, Yan Pang, “Shared Spotlight Meridian: Distributed Sparse Pseudorandom Functions for Scalable Federated Learning”, IEEE Symposium on Security and Privacy 2026 (S&P 2026), pp. 2269–2287, doi: 10.1109/SP63933.2026.00065. [\[Paper\]](#)
5. Peihua Mai, Youlong Ding, Ziyang Lyu, Minxin Du, Yan Pang*, “SecEmb: Sparsity-Aware Secure Federated Learning of On-Device Recommender System with Large Embedding”,

- Proceedings of the Forty-second International Conference on Machine Learning (ICML 2025), PMLR, 2025 [\[arXiv\]](#) [\[Paper\]](#)
6. Shuran Fu, Peihua Mai, Zhang Jingqi, Yan Pang*, “Utilizing Language Models for Synthetic Knowledge Graph Generation”, ICLR 2025 Workshop Data Problems, [\[Paper\]](#)
 7. Peihua Mai, Ran Yan, Yan Pang*, “RFLPA: A Robust Federated Learning Framework against Poisoning Attacks with Secure Aggregation”, Proceedings of the Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024), [\[arXiv\]](#)
 8. Fengzhu Zeng, Wenqian Li, Wei Gao, Yan Pang, “Multimodal Misinformation Detection by Learning from Synthetic Data with Multimodal LLMs”, Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP 2024), [\[arXiv\]](#)
 9. Peihua Mai, Ran Yan, Zhe Huang, Youjia Yang, Yan Pang*, “Split-and-Denoise: Protect large language model inference with local differential privacy”, Proceedings of the Forty-first International Conference on Machine Learning (ICML 2024), PMLR, 2024 [\[arXiv\]](#)
 10. Wenqian Li, Shuran Fu, Fengrui Zhang, Yan Pang*, “Data Valuation and Detections in Federated Learning”, published in the IEEE/CVF Conference on Computer Vision and Pattern Recognition 2024 (CVPR 2024) [\[arXiv\]](#)
 11. Wenqian Li, Shuran Fu, Yan Pang*, “Distributionally Robust Federated Learning with Wasserstein Barycenter”, The Twelfth International Conference on Learning Representations (ICLR 2024), TinyPaper [\[Paper\]](#)
 12. Peihua Mai, Hao Jiang, Ran Yan, Youjia Yang, Zhe Huang, Yan Pang*, “Knowledge Graph Unlearning to Defend Language Model Against Jailbreak Attack”, The Twelfth International Conference on Learning Representations (ICLR 2024), TinyPaper (Oral) [\[Paper\]](#)
 13. Peihua Mai and Yan Pang*, “Vertical Federated Graph Convolution Network for Recommender System”, Proceedings of the 40th International Conference on Machine Learning (ICML 2023), PMLR 202:23516-23535, 2023. [\[arXiv\]](#)
 14. Wenqian Li, Yinchuan Li, Zhigang Li, Jianye Hao, Yan Pang*, “DAG Matters! A Flow Network Enhanced Explainer for Graph Neural Networks”, Eleventh International Conference on Learning Representations (ICLR 2023) [\[arXiv\]](#)
 15. Yichu Xu, Wenqian Li, Yinchuan Li, Yunfeng Shao, Yan Pang, De-Chuan Zhan, “Two Important Things To Do Before Federated Training”, published in Eleventh International Conference on Learning Representations (ICLR 2023), TinyPaper
 16. Yan Pang*, Ganeshkumar Sundararaj, “Cryptocurrency price prediction using time series and social sentiment data”, 6th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies, Dec 2~4, 2019, Auckland, New Zealand
 17. Changhua Chen, Yan Pang*, “Exploring Machine Learning Techniques for Smart Drainage System”, 2019 IEEE Fifth International Conference on Big Data Computing Service and Applications (BigDataService) , Apr, 2019, San Francisco, CA, USA, pp.63-71. [\[Paper\]](#)
 18. Jun Sheng Lee, Denis Zuba, Yan Pang*, “Sentiment Analysis of Chinese Product Reviews using Gated Recurrent Unit”, 2019 IEEE Fifth International Conference on Big Data Computing Service and Applications (BigDataService) , Apr, 2019, San Francisco, CA, USA, pp.173-182. [\[Paper\]](#)
 19. Alwin Zhang, Sean Shao Wei Lam, Nan Liu, Yan Pang, Ling Ling Chan and Phua Hwee Tang, “Development of a Radiology Decision Support System for the Classification of MRI Brain Scans”, 5th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies, Dec, 2018, Zurich, Switzerland, (Best Student Paper Finalists)
 20. Yan Pang, M.L. Yuan, A.Y.C. Nee, S.K. Ong and Kamal Youcef-Toumi*, “A Markerless Registration Method for Augmented Reality based on Affine Properties”, Proceedings of the Seventh Australasian User Interface Conference (AUIC 2006), Hobart, Australia, CRPIT 50, W. Piekarski (ed.), pp. 25–32, 2006.
 21. Yan Pang, A.Y.C. Nee, S.K. Ong and Kamal Youcef-Toumi*, “Evaluation and Planning of Manual Assembly on Workstation in an Augmented Reality Environment”, Singapore-MIT Alliance Annual Symposium, 17-19 January 2006, Singapore

22. A.Y.C. Nee*, S.K. Ong, Yan Pang and J.W.S. Chong, 2006, “Robot Programming and Assembly Design using Augmented Reality”, Proceedings of the International Manufacturing Conference with China on Advances in Manufacturing Engineering, IMCC2006, ed. C.Y. Jiang, Northwestern Polytechnical University Press, ISBN 7-5612-2132-0, keynote address, 21-23 September 2006, Xian, Shaanxi, China, p. 4
23. Yan Pang, A.Y.C. Nee*, Kamal Youcef-Toumi, S.K. Ong and M.L. Yuan, “Assembly Design and Evaluation in an Augmented Reality Environment”, Singapore-MIT Alliance Annual Symposium, 19-20 January 2005, Singapore

Journal Papers (* Corresponding author)

24. Peihua Mai, Ran Yan, Youjia Yang, Rui Ye, Wei Liu, Yan Pang*, “ConfusionPrompt: Practical Private Inference for Online Large Language Models”, IEEE Transactions on Artificial Intelligence, 2026, doi: 10.1109/TAI.2026.3734014. [\[arXiv\]](#) [\[Paper\]](#)
25. Peihua Mai, Youjia Yang, Ran Yan, Yan Pang*, “Nexus Scissor: Enhance Open-Access Language Model Safety by Connection Pruning”, npj Artificial Intelligence 2, 1 (2026). [\[Paper\]](#)
26. Yan Pang*, Khor Ping Quek, “Bridging Academia and Industry in Business Analytics Education: Insights from the National University of Singapore”, INFORMS Journal on Applied Analytics 56(4), pp. 318–325, 2026, [\[Paper\]](#)
27. Peihua Mai and Yan Pang*, "Privacy-Preserving Multiview Matrix Factorization for Recommender Systems," in IEEE Transactions on Artificial Intelligence, vol. 5, no. 1, pp. 267-277, Jan. 2024, doi: 10.1109/TAI.2023.3240700. [\[arXiv\]](#) [\[Paper\]](#)
28. Yan Pang*, Zhen Yu, Haisheng Zhang, Corinne Sim, Marianne Louise Chang, Ashley Chong, “Blockchain Technology Transforms the Pharmaceutical Supply Chain”, Management and Business Review, vol. 5, no. 1, 2025, doi: 10.1177/2694104X251349034, [\[Paper\]](#) (**Zuellig Pharma launched a new product, eZTracker, based on this research: [\[Website\]](#)**)
29. Lin Du and Yan Pang*, “A novel data-driven methodology for influenza outbreak detection and prediction”, Sci Rep 11, 13275 (2021). [\[Paper\]](#) (**Zuellig Pharma launched a new product, Infection Watch Initiative, based on this research: [\[Website\]](#)**)
30. David De Cremer, Yan Pang, “Can Blockchain Manage Trust in Organizations?”, Management and Business Review, Vol. 1, No. 3, Fall 2021, pp. 83~90, [\[Paper\]](#)
31. Yan Pang*, “Empowering Small and Medium Enterprises through the Synergy of AI and Blockchain”, The European Business Review, Jan 2021, [\[Paper\]](#)
32. Yan Pang*, "A New Consensus Protocol for Blockchain Interoperability Architecture," in IEEE Access, vol. 8, pp. 153719-153730, 2020, doi: 10.1109/ACCESS.2020.3017549. [\[Paper\]](#)
33. Yu Zhen, Yan Pang*, Khor Ping Quek, Cheng Lin, “Geolocation Analysis Framework for Property Price Evaluation”, Journal of Real Estate Finance, Fall 2020, pp. 38~52
34. Yan Pang*, “How Big Data Analytics Impacts Evidence-based Healthcare”, Sigma: Insurance Research, Swiss Re Ltd, Aug, 2016
35. S.K. Ong, Yan Pang and A.Y.C. Nee*, “Augmented Reality Aided Assembly Design and Planning”, Annals of the CIRP, Vol. 56, No. 1, pp. 49–52, 2007, doi: 10.1016/j.cirp.2007.05.014. [\[Paper\]](#)
36. Yan Pang, A.Y.C. Nee, S.K. Ong, M.L. Yuan and Kamal Youcef-Toumi*, “Assembly Feature Design in an Augmented Reality Environment”, Assembly Automation, Vol.26, No.1, pp.34-43, January, 2006. [\[Paper\]](#)
37. Y.Q. Xia, R.Z. Tang*, and Yan Pang, “Research in Production Planning and Scheduling for Semi-process Industry”, Modular Machine Tool & Automatic Manufacturing Technique, 2003, No. 1, pp. 12 – 15

Company Papers (selected):

38. Lam Shao Wei, Teo Joo Keng, Chiew Koon Yeow, Rajagopal Mohanavalli, Yan Pang, Ong Eng Hock, “Predictive Analytics for the Waiting Times of Critical Patients in the Emergency Department”, SGH 21st Annual Scientific Meeting, Apr, 2015

39. Yan Pang*, Xin Sheng Mao, “Next Generation Information Technology”, chapter in the National Report of “China’s 12th Five-Year Development Plan”, published by China National Development and Reform Commission (NDRC), Nov, 2012
40. Yan Pang*, L. Lian, “An ontology-based Asset Portal for Harvesting Optimization Solution Assets”, IBM GCG RTE Conference, Aug, 2010 (**Best Paper Award**)
41. Yan Pang*, G. LAM, R. McLAIN, Y. Bai, C.L. Ren, “FAW Car Chooses Manufacturing Productivity Optimization Solution to Improve Production Productivity and Reduce Cost”, IBM IMPACT 2010, May, 2010
42. Yan Pang* and Y.S. Yip, “A Production Scheduling Model of Surface Mount Technology Lines”, 2008 Motorola Technical Symposium (2008MTS), Jul, 2008
43. Yan Pang*, “RFID in Supply Chain Operations: Opportunities and Challenges”, Jul, 2007, Motorola Technical Report
44. Yan Pang, “Design for Logistics”, Oct, 2007, Motorola Technical Report

Book (chapter):

45. Prashant Jain, Yan Pang, “Cloud Computing Architecture Patterns”, Chapter on “Handbook of Research on End-to-End Cloud Computing Architecture Design”, IGI Global Publisher, July 2016

Technology Column Articles:

I am a part-time tech columnist for Lianhe Zaobao (联合早报)([\[Website\]](#)) and Think China ([\[Website\]](#))

46. Yan Pang, “Breaking ground: China’s strides in AI regulation”, Think China, Oct, 2023, [\[Article\]](#)
47. Yan Pang, Yonggang Wen, “ChatGPT will permeate all facets of our work and life”, Think China, Apr, 2023, [\[Article\]](#)
48. Yan Pang, “How geopolitics will drastically change chip manufacturers’ ecosystem”, Jan, 2023, [\[Article\]](#);
49. 庞严, “中美竞争下世界半导体业何去何从? ”, 联合早报, Jan, 2023, [\[Article\]](#)
50. Yan Pang, X.J. Liang, “Metaverse: What is it and Who’s in control?”, Think China, Mar, 2022, [\[Article\]](#);
51. 庞严, 梁信军, “解析元宇宙及中美元宇宙发展现状”, 联合早报, May, 2022, [\[Article\]](#)
52. Yan Pang, “Who is winning the AI race: US or China”, Think China, Dec, 2021, [\[Article\]](#);
53. 庞严, “中美人工智能竞争分析”, 联合早报, Dec, 2021, [\[Article\]](#)
54. Yan Pang, “Why no country can win the chip war”, Think China, Aug, 2021, [\[Article\]](#)
55. Yan Pang, “China’s central bank digital currency (CBDC) innovations”, Think China, Jun, 2021, [\[Article\]](#);
56. 庞严, “解析中国央行数字货币及数码人民币”, 联合早报, Jul, 2021, [\[Article\]](#)
57. Y. Pang, “How China took the lead in the digital currency race”, Think China, Jun, 2021, [\[Article\]](#)
58. Yan Pang, “Has China surpassed the US in the 5G race?”, Think China, Mar, 2021, [\[Article\]](#);
59. 庞严, “5G 的中美之争及展望”, 联合早报, Apr, 2021, [\[Article\]](#)

International Industry Conference Presentations and Talks (selected)

60. *SAP APAC Executive Exchange for Discrete Industry*, Sept 2026, Singapore, (Topic: Trustworthy AI for the Autonomous Enterprise - Building Effective Human-Agent Collaboration)

61. *SAP Academic Community Conference APAC*, Jul 2025, Manila, Philippines, (Topic: Bridging the Gap between Industry and Academia – Best Practice from NUS MSBA Programme)
62. *SAP Academic Community Conference APJC*, Jul 2022, Melbourne, Australia, (Topic: How to Train the Next Generation Talents)
63. *SAP Global Analytics and Data Conference*, Jun, 2021 (Topic: The Synergy between Federated Learning and Blockchain - A New AI Learning Paradigm with Privacy Protection)
64. *China Blockchain and AI Forum*, Mar, 2019, Chongqing, China (Topic: The synergy between Big Data, AI and Blockchain)
65. *Singapore WOLOT Foundation Forum*, Singapore, Jan, 2019, Singapore (Topic: Cryptocurrency Value Evaluation Framework)
66. *China-ASEAN Digital Economy Summit*, Dec, 2018, Nanning, Guangxi, China, (Topic: The Frontiers in Artificial Intelligence)
67. *AACSB Asia Pacific Annual Conference*, Oct, 2018, Nagoya, Japan, (Topic: Incorporating Tech Trends into the Business Analytics Education)
68. *The 3rd China-ASEAN Information Harbor Forum*, Sep, 2018, Nanning Guangxi, China, (Topic: The Synergy of AI and BlockChain)
69. *China BlockChain Innovation and Application Associate Conference*, Jun, 2018 Chongqing, China, (Topic: Singapore FinTech and BlockChain Strategy and Landscape)
70. *Internet Thinker Conference*, Nov, 2017, Singapore, (Topic: Rethink of Artificial Intelligence)
71. *The 3rd Global Healthcare Big Data Conference*, Nov 9~11, 2017, Shenzhen, China, (Topic: Healthcare Analytics Key Technologies)
72. *2017 Global Big Data for Healthcare Workshop*, Peking University, Jan, 2017, Beijing, China, (Topic: Healthcare Operations Analytics)
73. *Europe Data Science Conference (EDSC) 2016*, Nov, 2016, Luxembourg, (Keynote: Big Data and Business Analytics: Accelerating Digital Transformation in Enterprises and Industries)
74. *China Smart Enterprise Summit 2016*, Sep, 2016, Boao, Hainan, China, (Topic: Big Data Analytics and Optimization in Supply Chain Planning)
75. *Big Data for Precision Medicine Conference*, The University of Hong Kong, Jun, 2016, Hong Kong, China, (Topic: A Study of Personalized Patient Treatment in Rheumatoid Arthritis (RA))

Media Coverage (selected)

76. Channel News Asia, Jun 2026, (Topic: The rise of the AI ‘token economy’ and China’s big bet on scale), <https://www.channelnewsasia.com/east-asia/ai-tokens-token-economy-china-6159196>
77. Lianhe Zaobao (联合早报), May 2026, (Topic: 从马斯克对决奥特曼：看万亿估值下的 AI 治理之争) https://www.zaobao.com.sg/news/world/story20260522-9054566?utm_source=android-share&utm_medium=app
78. Channel News Asia, May, 2025, (Topic: State of Artificial Intelligence) <https://www.youtube.com/watch?v=cahzQaEao2s>
79. Lianhe Zaobao (联合早报), Feb 2025, (Topic: 中国应对美国关税战推动本土科技产业崛起) <https://www.zaobao.com/news/china/story20250221-5883766>
80. *Singapore MediaCorp Channel 8 晨光聚焦*, Jan, 2025, (Topic: Crypto Market Insights and Projection), <https://www.8world.com/in-depth/morning-express/spotlight-crypto-investment-trend-2694706>

81. Lianhe Zaobao (联合早报), Nov 2024, (Topic: 学者主张人工智须先治理后发展 业界呼吁“先让子弹飞”) <https://www.zaobao.com.sg/news/china/story20241111-5281779>
82. Lianhe Zaobao (联合早报), Jul 2024, (Topic: 中国 AI 行业“大洗牌” 迈向本土化) <https://www.zaobao.com.sg/news/china/story20240707-4074279>
83. Singapore MediaCorp Channel 8 狮城有约, Oct 2023, (Topic: Singapore digital economy report), <https://www.8world.com/stories/hello-singapore/straight-talk-digital-economy-development-2260686>
84. HEALTHCAREASIA, Jul 2023, (Topic: Vendor View: Adopting AI in healthcare system? Build data frameworks first!) <https://healthcareasiamagazine.com/healthcare/exclusive/vendor-view-adopting-ai-in-healthcare-system-build-data-frameworks-first>
85. Lianhe Zaobao (联合早报), Jun 2023, (Topic: 北上深接连发布支持 AI 发展政策的解读) <https://www.zaobao.com/news/china/story20230601-1400406>
86. Lianhe Zaobao (联合早报), Mar 2023, (Topic: 中美科技巨头人工智能商海争霸 谁家机器人最能聊谁是赢家), <https://www.zaobao.com.sg/news/world/story20230305-1369326>
87. Lianhe Zaobao (联合早报), Mar 2023, (Topic: 疫情加快数码转型进度 人工智能启动企业新引擎), <https://www.zaobao.com.sg/finance/singapore/story20230108-1351139>
88. The Straits Times, Feb, 2023, (Topic: Will AI take over people's jobs and is Singapore prepared?), <https://www.straitstimes.com/business/will-ai-take-over-peoples-jobs-and-is-singapore-prepared>
89. Workipedia MyCareerFutures, Oct 2022, (Topic: Working in the “Metaverse”: What Jobs Could Arise and What Skills Would Singaporean Workers Need to Log in?), <https://content.mycareersfuture.gov.sg/working-metaverse-what-jobs-arise-what-skills-singaporean-workers-need-log/>
90. CNA938 Tech Talk, Aug, 2022, (Topic: Non-Fungible Token (NFT) hype)
91. Singapore MediaCorp Channel 8 十分访谈, May 2022, (Topic: Why Terra Luna and UST crash?) <https://www.8world.com/stories/hello-singapore/straight-talk-risk-of-stable-coins-and-cryptocurrencies-1813846>
92. Singapore MediaCorp Channel 8 狮城有约, Dec, 2021, (Topic: Real Estate Investment in Metaverse) <https://www.8world.com/stories/hello-singapore/straight-talk-metaverse-virtual-property-1669956>
93. Xinhua (新华社), Nov 2021, (Topic: The impact of China's application to join the Digital Economy Partnership Agreement (DEPA)) <http://world.people.com.cn/n1/2021/1104/c1002-32273762.html>
94. Lianhe Zaobao (联合早报), 中新建交 30 特别论坛, Dec 2020, (Topic: China tech industry present and future outlook) <https://www.zaobao.com.sg/znews/singapore/story20201218-1109625>
95. CNA (Channel News Asia), Dec, 2020, (Topic: Impact of DBS to launch world's first crypto exchange backed by a traditional bank) <https://www.youtube.com/watch?v=JtG6h94-wWg>
96. China Business Network (第一财经), Mar 2020, (Topic: COVID-19 impact on global supply chain)
97. Singapore MediaCorp Capital 958, Jan 2020, (Topic: Singapore Digital Bank License and its impact in the financial industry)

98. 21st Century Business Herald (21 世纪经济报道), Jan 2020, (Topic: The Cloud Computing Technology and Market Development in South East Asia)
99. *Singapore MediaCorp Channel 8* 财经追击, Jul 2019, (Topic: The impact of Facebook Libra in the global finance system)
100. CCTV (China Central Television) securities information channel (中国中央电视台财经证券资讯频道), May 2019, (Topic: Focus on BlockChain 聚焦区块链)
101. Xinhua Finance, Dec 2018, (Topic: Why can Singapore become a global blockchain hub?)
102. Singapore MediaCorp Capital 958, Oct 2014, (Topic: IBM Watson AI technology and its application in education)

Company Invited Talks (selected)

103. *SAP AI Community Forum*, Aug, 2022 (Topic: Data Scientist Skills for the Future)
104. *Julius Baer Blockchain Innovation Forum*, May, 2022 (Topic: Blockchain and Cryptocurrency – Hype Or Trend?)
105. *Huawei Seeds For The Future Conference*, Singapore, Dec, 2021 (Topic: Skills for the Future)
106. *ZhongXin (Chongqing) Enterprise Forum*, Chongqing, China, Nov, 2021 (Topic: Industry 4.0 and Digital Transformation)
107. *PSA Data Festival*, Singapore, Sep, 2020 (Topic: Synergy Between Federated Learning and Blockchain)
108. *Dyson Global Data Science Annual Conference*, Singapore, Sep, 2019 (Topic: Business Analytics Landscape and Future Outlook)
109. *Johnson & Johnson Advanced Analytics Seminar*, Singapore, Aug, 2019 (Topic: Co-Innovations between Industry and Academia in Business Analytics)
110. *BYD Smart Manufacturing Seminar*, Shenzhen, China, May, 2019 (Topic: Optimization-based Production Planning and Scheduling in Automotive Industry)
111. *DELL Technical Forum*, Singapore, Oct, 2018 (Topic: The Smart Supply Chain)
112. *IBD Technology Technical Forum*, Singapore, Sep 2018, (Topic: BlockChain Transforms Traditional Loyalty Programs)
113. *Zhejiang University Research Seminar*, Hangzhou, Zhejiang, China, Jun, 2018 (Topic: Business Analytics: Progress on Applications and Future Outlook)
114. *FAW Manufacturing System Workshop*, Changchun, Jilin, China, Jun, 2018 (Topic: Industry Big Data Analytics and Optimization Platform)
115. *SingHealth Health Services Research Centre Seminar Series*, SingHealth, Singapore, May, 2018 (Topic: Big Data Analytics Transforms Healthcare Industry)
116. *AXA Technology forum*, Jun, 2017, Singapore, (Topic: business Transformation under ABCD Technologies)
117. *HuaWei Enterprise Digitalization – Operations Transformation Global Conference*, Singapore, 24 May, 2017 (Panel Discussion Member, Topic: Digital Transformation in Global Enterprise)
118. *IHiS Healthcare Technology Seminar*, Integrated Healthcare Information Systems Pte Ltd, Singapore, Jun, 2016 (Topic: Key Technologies in Healthcare Analytics)
119. *Microsoft Singapore Data Scientist Seminar*, Singapore, May, 2015 (Topic: Fundamentals in Business Analytics)
120. *IBM GMU Technical Conference*, Dubai, UAE, Feb, 2014 (Topic: Healthcare Analytics Reference Architecture)
121. *Singapore Healthcare Analytics Workshop*, IBM, Singapore, Nov, 2013 (Topic: The Next Frontier in Healthcare Analytics)

122. *Social Program Management Solution Technical Workshop*, Ministry of Social and Family Development, Singapore, Sep, 2013 (Topic: Analytics in Next Generation Social Service Platform)
123. *IRAS Technical Workshop*, Inland Revenue Authority of Singapore, Singapore Aug, 2013 (Topic: Apply Watson Cognitive Computing Technology in Citizen Service)
124. *IBM Healthcare Architect Annual Meeting*, Dallas, TX, US, Jun, 2013 (Topic: Healthcare Process Improvement – A Case Study from Singapore)
125. *Customs Solution Technical Workshop*, Singapore Customs, Singapore, Jan, 2013 (Topic: The Vision and Architecture Design of New TradeNet and TradeXchange)
126. *SuNing e-Commerce Symposium*, SuNing, Nanjing, Jiangsu, China, Jun, 2012 (Topic: Supply Chain Network Optimization and Inventory Optimization)
127. *IBM China Research and Development Open Day*, Beijiing, China, May, 2012 (Topic: Optimization-based Production Planning and Scheduling)
128. *Alibaba Cloud Computing Seminar*, Hangzhou, China, Mar, 2012 (Topic: Cloud Application Platform and API Economy)
129. *Advanced Manufacturing Technical Workshop*, Foton Motor, Beijing, China, Apr 2011 (Topic: Manufacturing Productivity Optimization Solution)
130. *LARC Research Seminar*, Singapore Management University, Singapore, Mar, 2011 (Topic: Advanced Analytics and Optimization (AAO) Framework to Solve Complex Real World Problems)
131. *Supply Chain Management Forum*, Tsingdao Brewery, Qingdao, China, Aug, 2010 (Topic: Production Planning and Scheduling in Process Manufacturing Industry)
132. *FAW production planning and scheduling seminar*, First Automotive Works (FAW), Changchun, China, Jan, 2010 (Topic: Improve Productivity using Optimization-based Production Planning and Scheduling)
133. *Manufacturing Technology Workshop*, Changan Ford, Chongqing, China, Sep, 2009 (Topic: How to Apply Optimization Techniques in Manufacturing System)
134. *Chery Automotive Technical Workshop*, Chery Automotive, Wuhu, Anhui, China, Aug, 2009 (Topic: Supply Chain Optimization)
135. *Manufacturing Technology Forum*, Motorola, Penang, Malaysia, May, 2008 (Topic: The Optimization of Printed Circuit Board Assembly Lines)

RESEARCH GRANT AND INDUSTRY COLLABORATION

Research Grants

1. Principal Investigator, S\$500,000 (Industry donation), Multi-Agent System, 2026 ~ 2030
2. Principle Investigator, S\$400,000, Research on Evaluation Methodology for Large Language Model (LLM), NUS (Chongqing) Research Institute, 2026~2029
3. Principle Investigator, S\$400,000, Research on Data Valuation and Selection for Model Training, NUS (Chongqing) Research Institute, 2026~2029
4. Principal Investigator, S\$1,800,000, Trustworthy Large Language Model, NUS Guangzhou Research Translation and Innovation Institute, 2024 ~ 2029
5. Principal Investigator, S\$200,000, Privacy and Security Technology in Large Language Model, National Nature Science Foundation of Chongqing, 2024 ~ 2026
6. Principal Investigator, S\$600,000, Synthetic Data Generation, PSR Technology, Jun 2023 ~ May 2025
7. Principal Investigator, S\$200,000, Cross-domain Recommender System, Corp IT and Singapore Data Science Consortium (SDSC), Jul 2021 ~ Jun 2022
8. Co-Principal Investigator, S\$300,000, Real-Time Misinformation Assessment and Propagation Analysis with Graph Neural Networks, Mar 2021 ~ Mar 2024

9. Principal Investigator, S\$540,000, Supply chain and production system optimization, FAW Car Pte Ltd, June 2019 ~ May 2021
10. Principal Investigator, S\$300,000, Company risk profiling using natural language processing and machine learning techniques, UOB Bank and Singapore Data Science Consortium (SDSC), May 2019 ~ Apr 2020
11. Principal Investigator, Beauty Trend Engine, S\$300,000, Meiyume Ptd Ltd and SDSC, Feb 2019 ~ Jan 2021.
12. Principal Investigator, Workforce Planning and Scheduling Optimization, S\$220,000, Jurong Port Pte Ltd and SDSC, Jan 2019 ~ Dec 2020.
13. Principal Investigator, Big Data Analytics Platform for Decision Making Support in New Industrial City (NIC), S\$160,000, China Fortune Land Development (CFLD) International, Jan 2018 ~ Dec 2020.

TEACHING AND STUDENT MENTORING

Teaching

Topics: Trustworthy AI, Explainable AI, Blockchain Technology and Applications, Digital Transformation, AI Native Organization

Student Mentoring and Awards

International Activity Mentoring:

<i>Students</i>	<i>Competition</i>	<i>Awards & Date</i>
Bhargav Sagiraju Soumya Haridas Rickson Hu	Huawei Developer Competition 2025 (APAC)	2 nd Runner-up, Nov, 2025 with prize of USD 2,000 (https://en.prnasia.com/releases/apac/huawei-developer-competition-2025-asia-pacific-grand-finals-successfully-concluded-513133.shtml)
Zhen Yu Xhoni Shollaj Ran Yan Junhui Xu	Huawei Cloud Developer Competition APAJ Final 2022	1 st Runner-up, Sep, 2022 with prize of USD 5,000 (https://bizbeat.nus.edu.sg/community-news/article/msba-team-shine-in-huawei-developer-competition-2022/)
Xhoni Shollaj Widya Salim Rachel Sng Gino Martelli Tiu	AIxImpact International Case Competition	1 st place, Feb, 2022 (https://bizbeat.nus.edu.sg/community-news/article/msba-team-wins-international-ai-case-competition/)
Team 2023 Yixin Sun, Kang Jie Ang,	Chengdu80 FinTech and	2 nd Prize in 2019 with prize of SGD 6,000

Shane Yuhe Xin, Anshumaan Phukan, Venkata Durga Akhil, Yin Joe Ng Team 2020 Alexander Baesell Tommy Kangdra Ken Cheah Yu Wenxi Liow Jia Chen Team 2019 Liu Tong Hua Xiuping Jonathan Simon Wagner Chen Yanzhu Solimine Peter Dylan Park Jinseo Team 2018 Shivam Bansal Yi Suqin Ganeshkumar Sundararaj Steven Tseng He Shiyuan Sung Zheng Jie	Analytics competition (participating universities include University of California Berkeley, National University of Singapore, Georgia Institute of Technology, University of Toronto, University of Zurich, Peking University, Tsinghua University, University of Hong Kong, Shanghai Jiaotong University, and SWUFE)	(https://msba.nus.edu.sg/news/nus-msba-students-clinches-first-runner-up-at-fintech80-chengdu-hackathon/) 1st Prize in 2020 with prize of SGD 8,000 (https://msba.nus.edu.sg/news/msba-students-win-chengdu-80-fintech-competition-2020/) 2nd Prize in 2019 with prize of SGD 6,000 (https://msba.nus.edu.sg/news/msba-students-first-runners-up-chengdu-fintech-comp-2019/) 1st Prize in 2018 with prize of SGD 8,000 (https://www.comp.nus.edu.sg/news/archives/y2018/2792-2018-chengdu-80-competition/)
Shivam Bansal	Kaggle City of Los Angeles workforce analytics competition	1st Prize with SGD 5,000 Apr 2019 (https://www.kaggle.com/c/data-science-for-good-city-of-los-angeles/discussion/98155)

Student Startups Mentoring

<i>Students</i>	<i>Startup (Mentoring Period)</i>	<i>Highlights</i>
Gong He	FingerDance (2021-2022)	FingerDance has been selected for NUS GRIP (Graduate Research Innovation Programme) round 8 incubation fund support in May 2022.

		FingerDance “See Your Voice” solution won 2022's Huawei Tech4City Finals with cash prize of S\$10,000. (https://www.straitstimes.com/tech/smart-glove-that-translates-sign-language-into-english-chinese-text-wins-contest)
Uzair Javaid	Betterdata (2021-2022)	Betterdata has raised a seed-fund with S\$500k. (https://www.betterdata.ai/)
Jake Goh Davis Gay Jing Rong Lim	RateX (2017)	RateX successfully raised S\$ 3million in pre-series A funding round. (https://www.finextra.com/pressarticle/72914/browser-based-payments-startup-ratex-raises-3-million)
Nicholas Ooi Joshua Foong Jane Lee	Bantu (2018-2019)	Bantu won President’s Challenge Social Enterprise Award (PCSEA) (Social Enterprise startup of the year) in 2019 President’s Challenge. (https://www.raise.sg/president-s-challenge-social-enterprise-award.html)

CONSULTATIONS (Alphabetical Order)

- Alibaba (Big data analytics, AI and Cloud Computing)
- AXA (Big data analytics and AI)
- Baidu (Solution Architecture, Analytics and Optimization)
- Bank of QingDao (Solution Architecture, Analytics and Optimization)
- China Fortune Land Development (CFLD) Co. (Geolocation Analytics)
- Changan Ford Automobile 长安福特 (Production Planning and Scheduling)
- Chery Automobile (Production Planning and Scheduling)
- China Foreign Exchange Trade System (CFETS) (Asset Portfolio Optimization)
- DACHSER (Production Planning and Scheduling)
- DBS (Big data analytics and AI)
- DELL (Supply Chain Analytics and Optimization)
- DHL (Supply Chain Analytics and Optimization)
- Dongfeng Motor Corporation (Production Planning and Scheduling)
- eBeeCare (Analytics and Optimization)
- FAW Car 中国一汽 (Supply Chain and Production Planning and Scheduling)
- GAC Group 广汽集团 (Supply Chain and Production Planning and Scheduling)
- Hongkong International Terminals (Yard Planning Optimization)
- Huawei (Big data analytics, AI and Cloud Computing)
- IBD Technology (Blockchain)
- HSBC (Gen AI)
- Integrated Health Information System (IHIS) (Big data analytics and AI)
- Inland Revenue Authority of Singapore (IRAS) (AI Chatbot)
- Jinvovo Technology Group (Big data analytics, AI and Blockchain)
- Johnson & Johnson (Big data analytics and AI)

- Jurong Port (workforce planning and scheduling)
- Luxembourg Institute of Science and Technology (LIST) (Big data and AI)
- Marina Bay Sands (Analytics and Optimization)
- Meiyume (Big data Analytics and AI)
- Microsoft (Analytics and Optimization)
- Ministry of Health Holding (MOHH) (Big data analytics and AI)
- Ministry of Social and Family Development (MSF) (Analytics and Optimization)
- Neptune Orient Lines (NOL) (Empty Container Relocation Optimization)
- RICHTEK (Supply Chain Planning)
- SAP (Big data analytics and AI)
- Shanghai Electric Group (Production Line Planning and Scheduling)
- Shanghai GM Automobile 上汽通用 (Production Line Planning and Scheduling)
- Singapore Customs (Analytics and Cloud Computing)
- Singapore Tourism Board (big data analytics and AI)
- SingHealth (Big data analytics and AI)
- State Street Corporation (Asset Analytics and Optimization)
- Swiss Re (Big data analytics and AI Strategy)
- Tetra Pak (Big data analytics and AI)
- Tigergraph (Analytics and AI)
- United Overseas Bank (UOB) (Analytics and AI)